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OpenShift Platform Overview

Virtualization, AI, and Developer Toolchain

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Agenda

- Why OpenShift as a unified platform
- OpenShift Virtualization – VMs and Kubernetes together
- OpenShift AI – ML and GenAI on the cluster
- Developer toolchain – from code to production
- How the pieces fit together
- Summary

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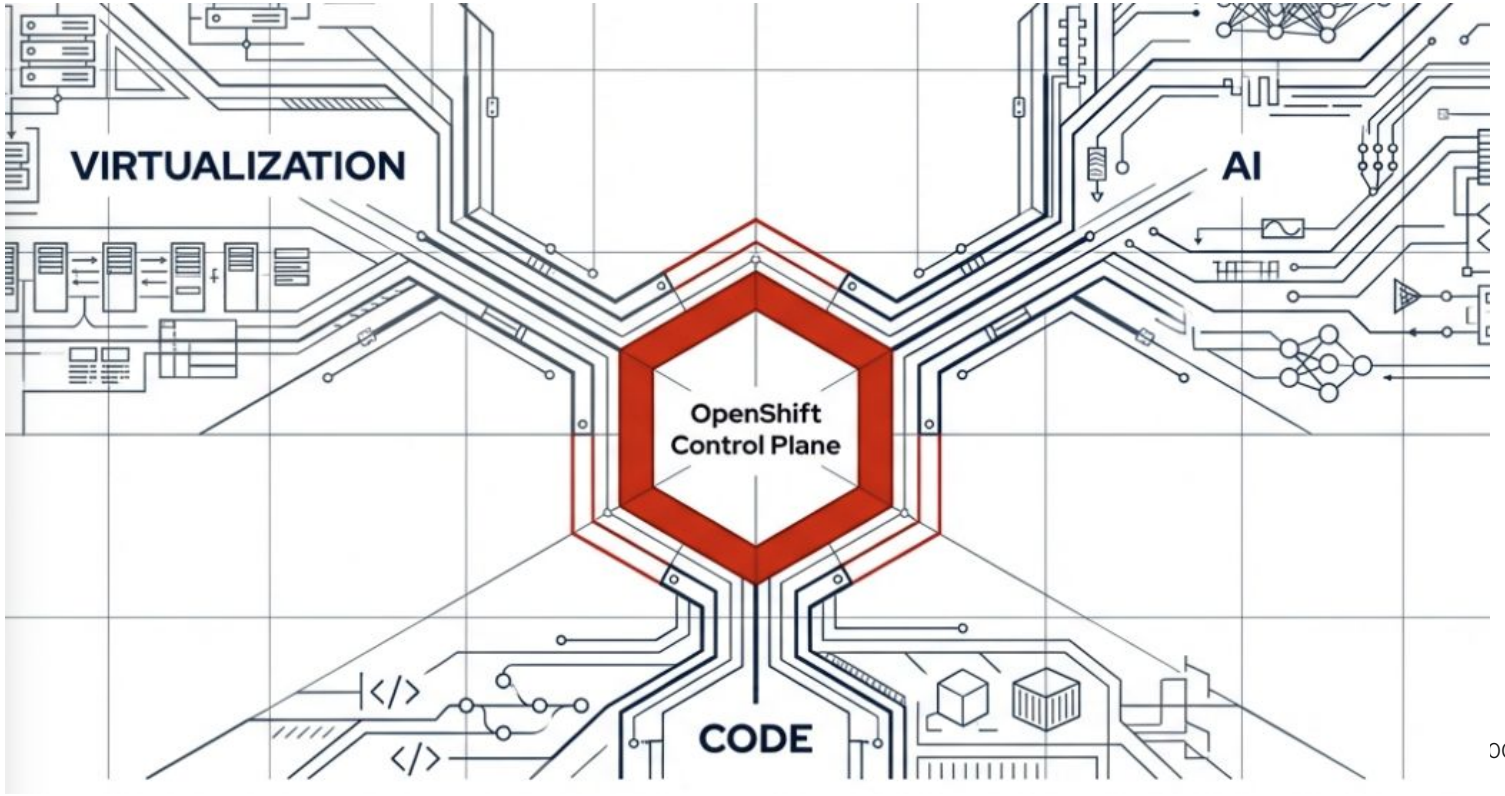
Unified Platform



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One Platform, Many Workloads

Containers at the core

- OpenShift extends Kubernetes with enterprise operations and security
- Same cluster can run cloud-native apps, VMs, and AI workloads
- Consistent RBAC, networking, storage, and GitOps across tiers
- Operators and add-ons install as managed components
- Red Hat support and lifecycle across the stack

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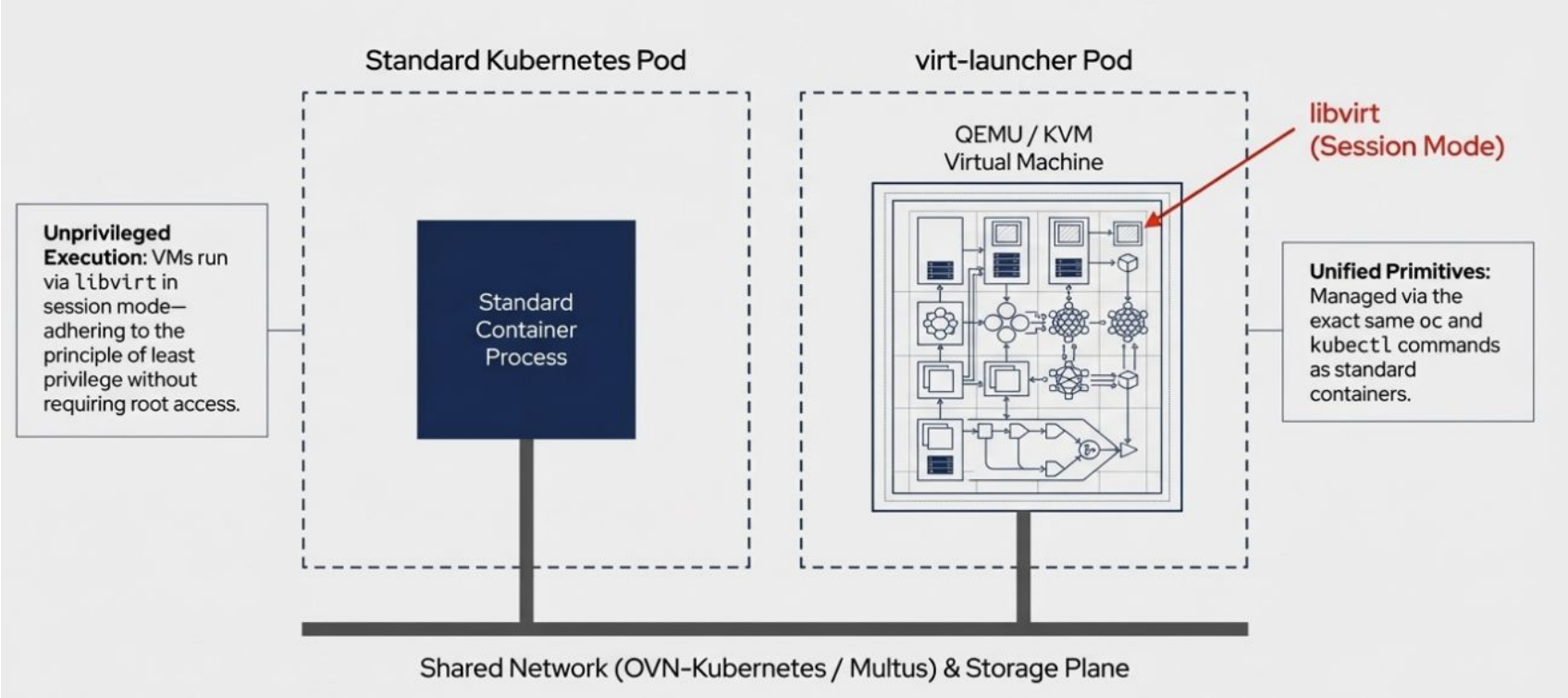
OpenShift Virtualization



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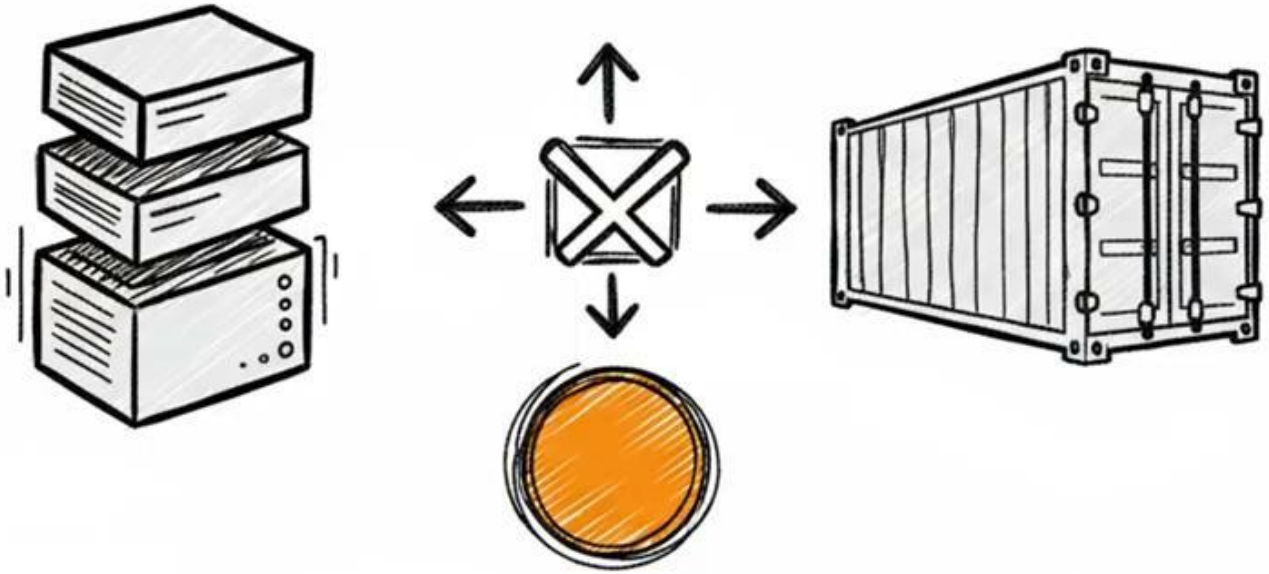
OpenShift Virtualization



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What Is OpenShift Virtualization

KubeVirt-based

- Runs virtual machines alongside containers on the same cluster
- Built on KubeVirt: VMs as first-class Kubernetes resources
- Manage VMs with oc, console, or GitOps like any workload
- Live migration, snapshots, and storage integration with ODF or CSI
- Bridge legacy apps and modern apps without two separate platforms

KubeVirt

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Virtualization – Typical Use Cases

- Lift-and-shift workloads that still need a full OS
- Mixed environments: new microservices next to existing VMs
- Development and test VMs on demand
- Gradual modernization: VM today, containerize later
- Single operational model: same monitoring, backup, and policies

KubeVirt

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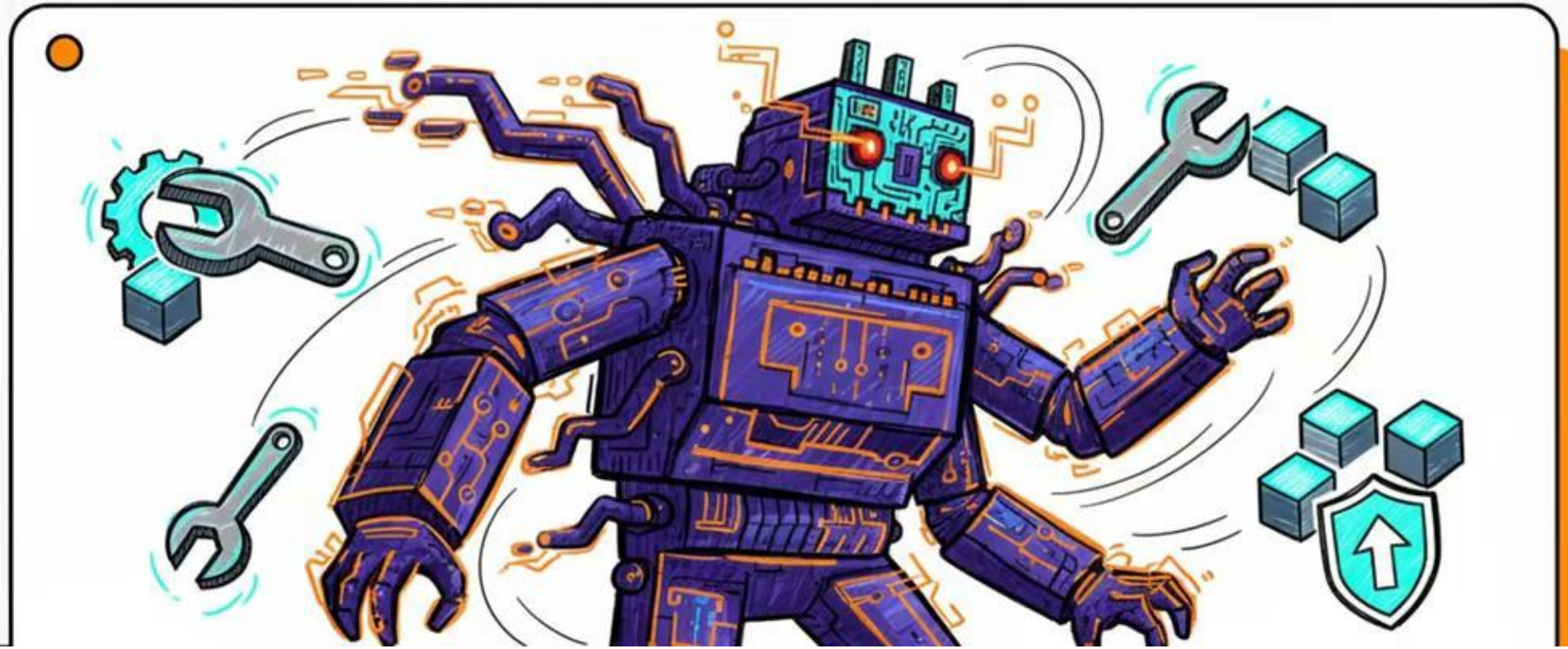
OpenShift AI

Taming the Titans



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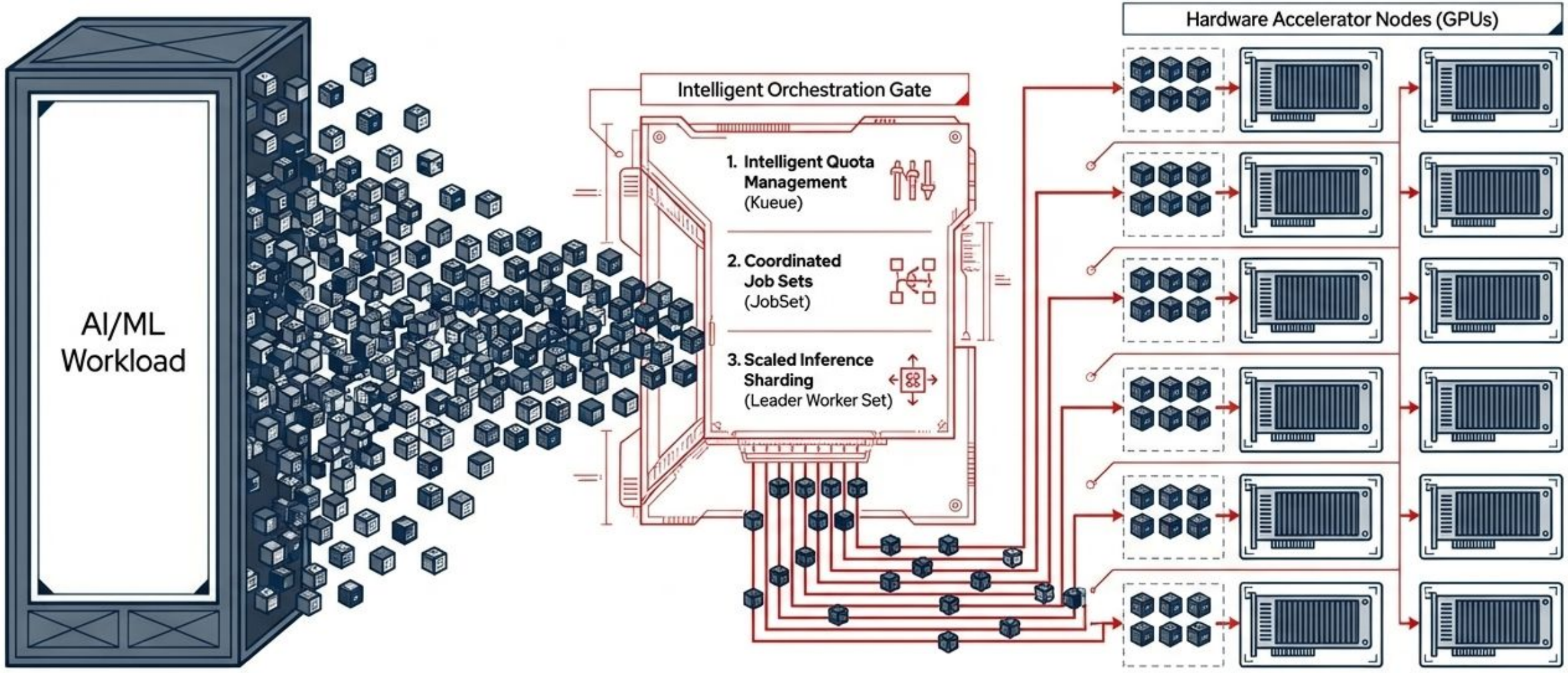
What Is OpenShift AI

Red Hat OpenShift AI

- Platform for data science, ML, and generative AI workflows
- Integrates notebooks, model serving, pipelines, and MLOps patterns
- Runs on OpenShift with GPU support where hardware allows
- Connects to enterprise data and governance requirements
- Aligns with hybrid cloud: train and serve where data lives

Orchestrating the Next Generation of AI Workloads

From Traditional Compute to Massive, Coordinated Distributed Training and Inference.



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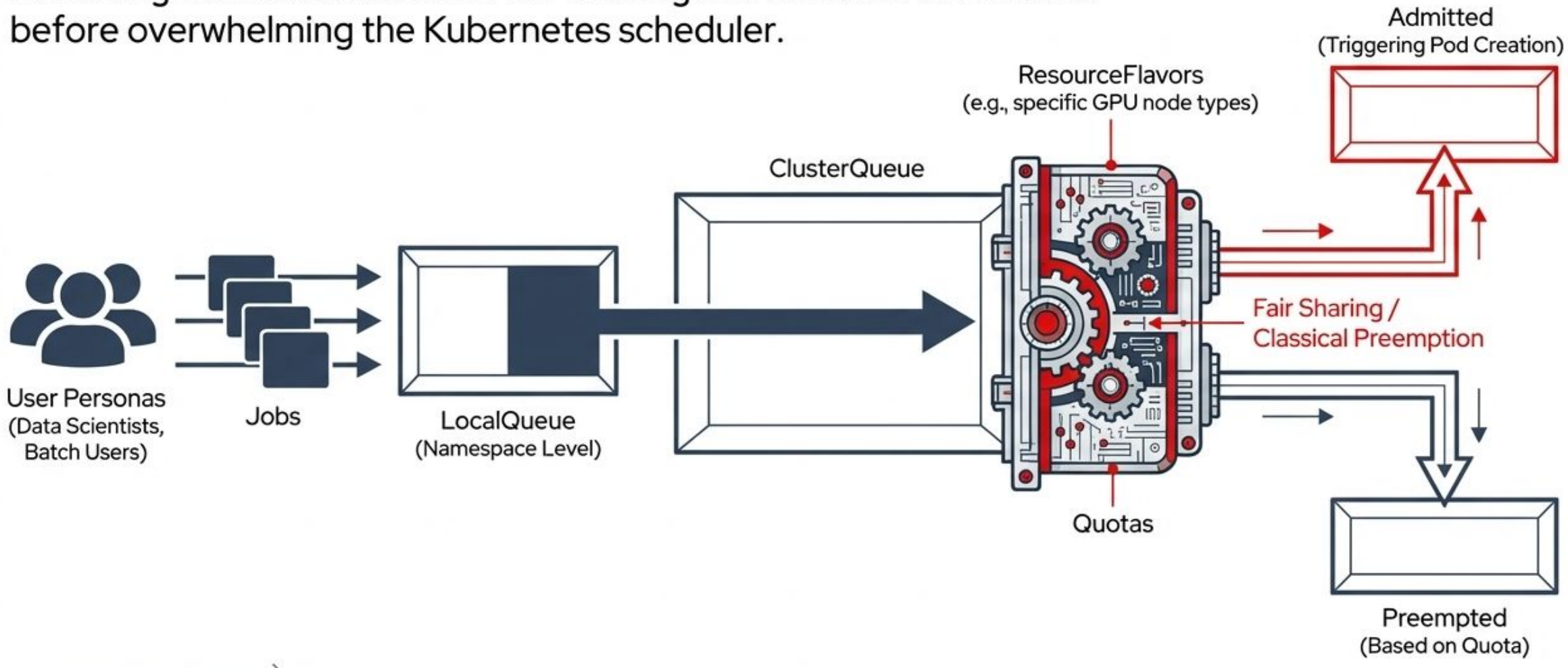
Red Hat build of Kueue: Intelligent Quota & Job Scheduling

Admitting workloads based on fair-sharing and resource constraints before overwhelming the Kubernetes scheduler.



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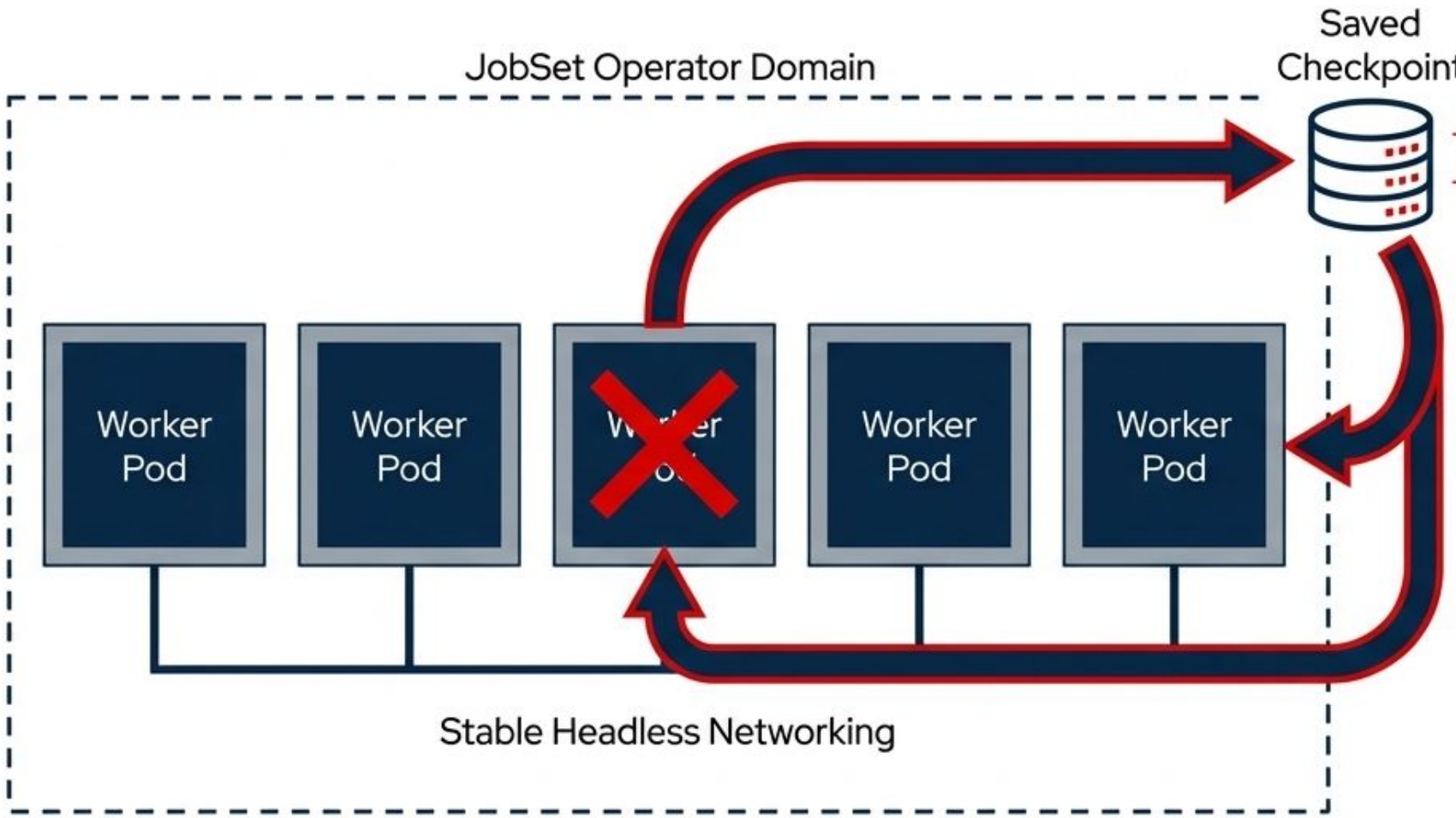
JobSet Operator: Coordinated Distributed Training



Group as a Unit: Manages many individual components as a single resilient system

Automatic Failure Recovery: Restarts the group from a saved checkpoint if one node fails

Stable Networking: Predictable IP addressing so workers can rediscover each other

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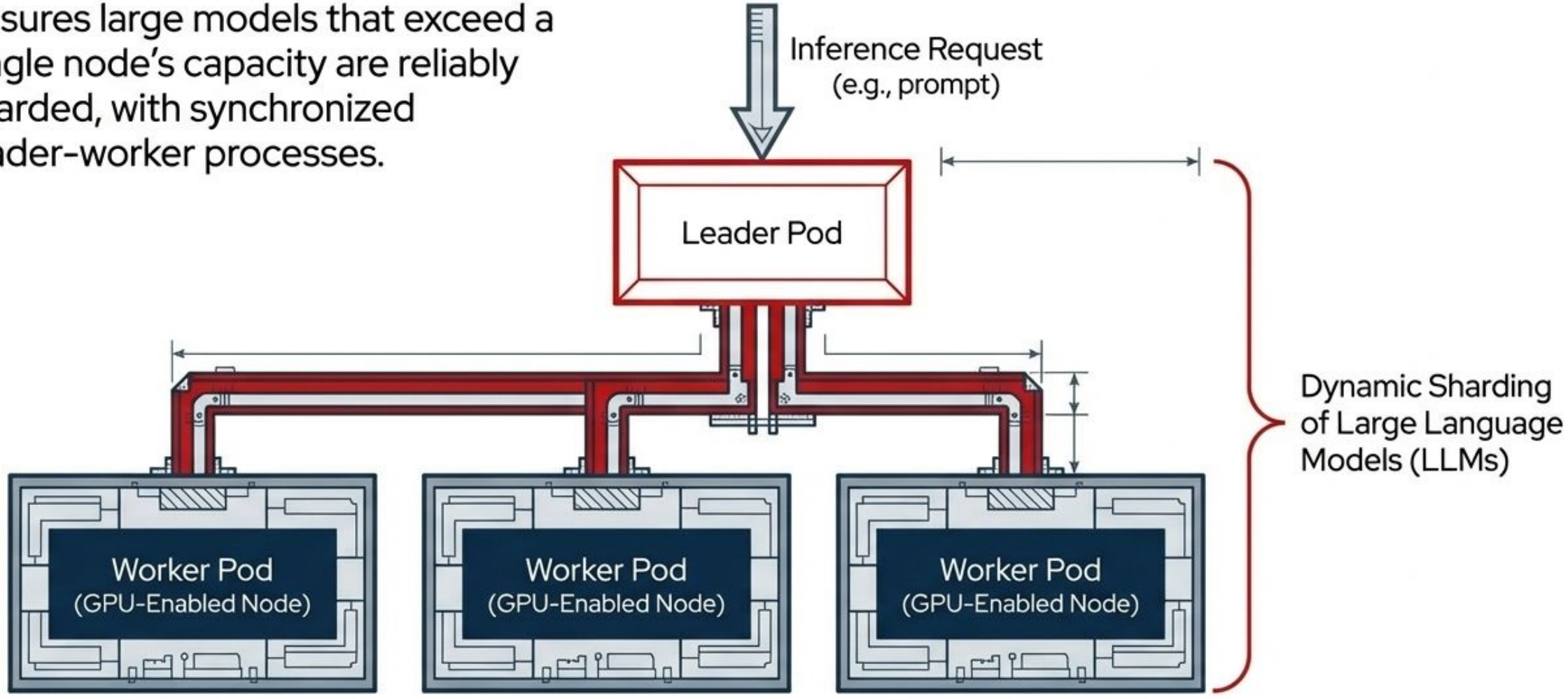
Leader Worker Set Operator: Scaling AI Inference

Ensures large models that exceed a single node's capacity are reliably sharded, with synchronized leader-worker processes.



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OpenShift AI – Typical Use Cases

- Self-service environments for data scientists and ML engineers
- Serving models behind OpenShift Routes and service mesh
- Integration with CI/CD for model promotion and testing
- Governance: namespaces, quotas, and audit for AI projects
- Hybrid: same patterns on cloud and on-prem clusters

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Developer Toolchain

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Developer Experience on OpenShift

Toolchain overview

- Web console and Developer perspective: deploy from Git, catalog, pipelines
- oc CLI and kubectl-compatible workflows
- OpenShift Pipelines (Tekton) for CI/CD on the platform
- Builds: Source-to-Image (S2I), BuildConfigs, or external CI
- GitOps: OpenShift GitOps (Argo CD) for declarative delivery

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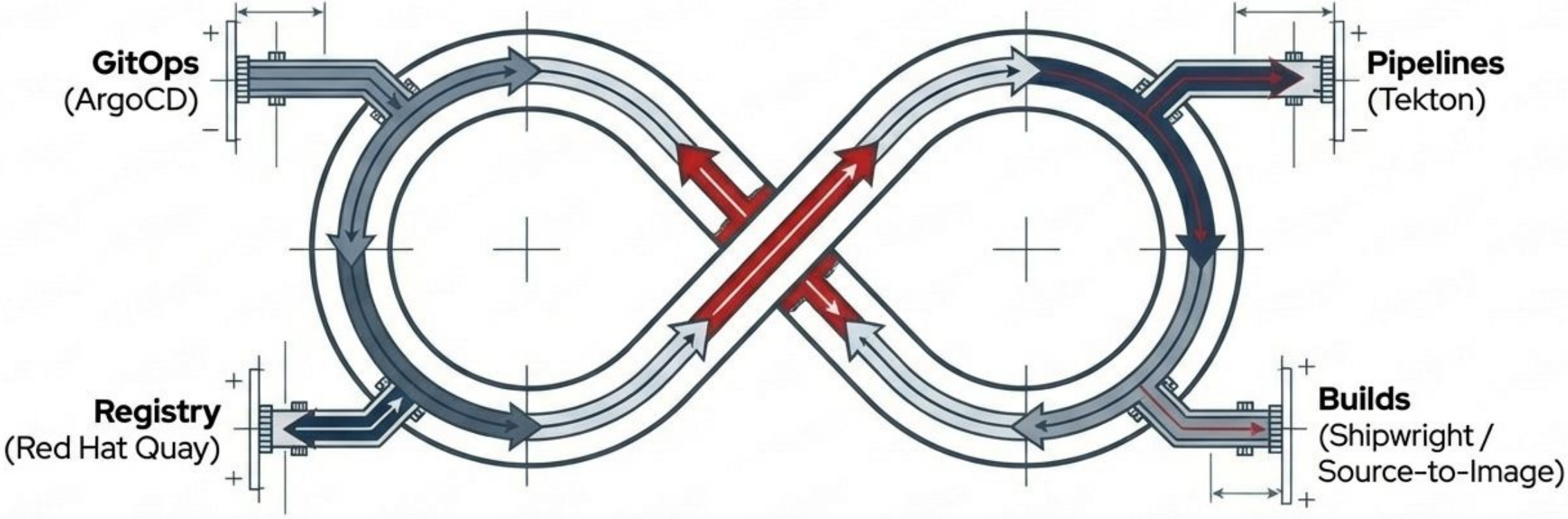
Accelerating the Path to Production

A Native, Frictionless Developer Toolchain

Treating infrastructure, deployments, and pipelines as version-controlled code.



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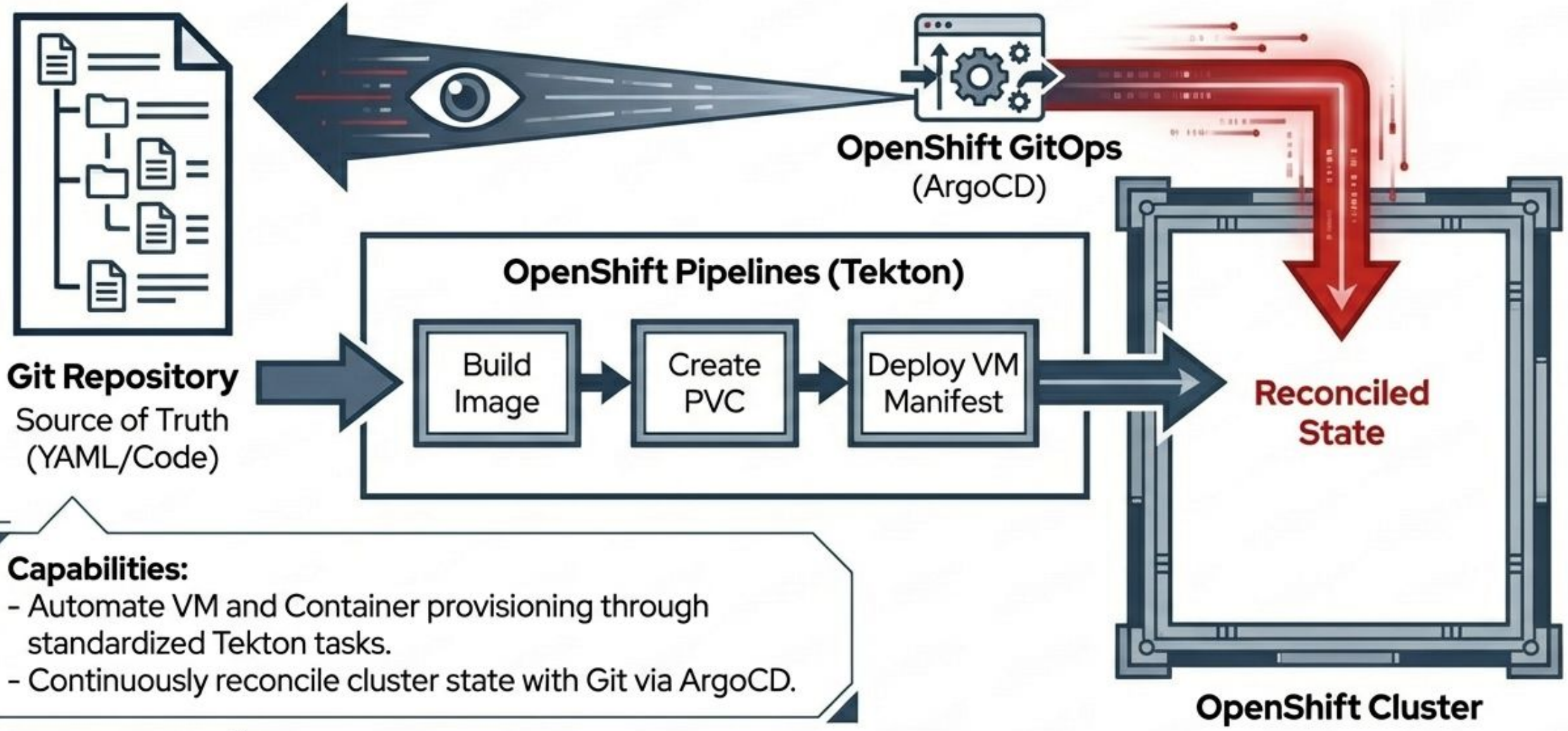
Declarative Automation: GitOps & OpenShift Pipelines



Capabilities:

- Automate VM and Container provisioning through standardized Tekton tasks.
- Continuously reconcile cluster state with Git via ArgoCD.

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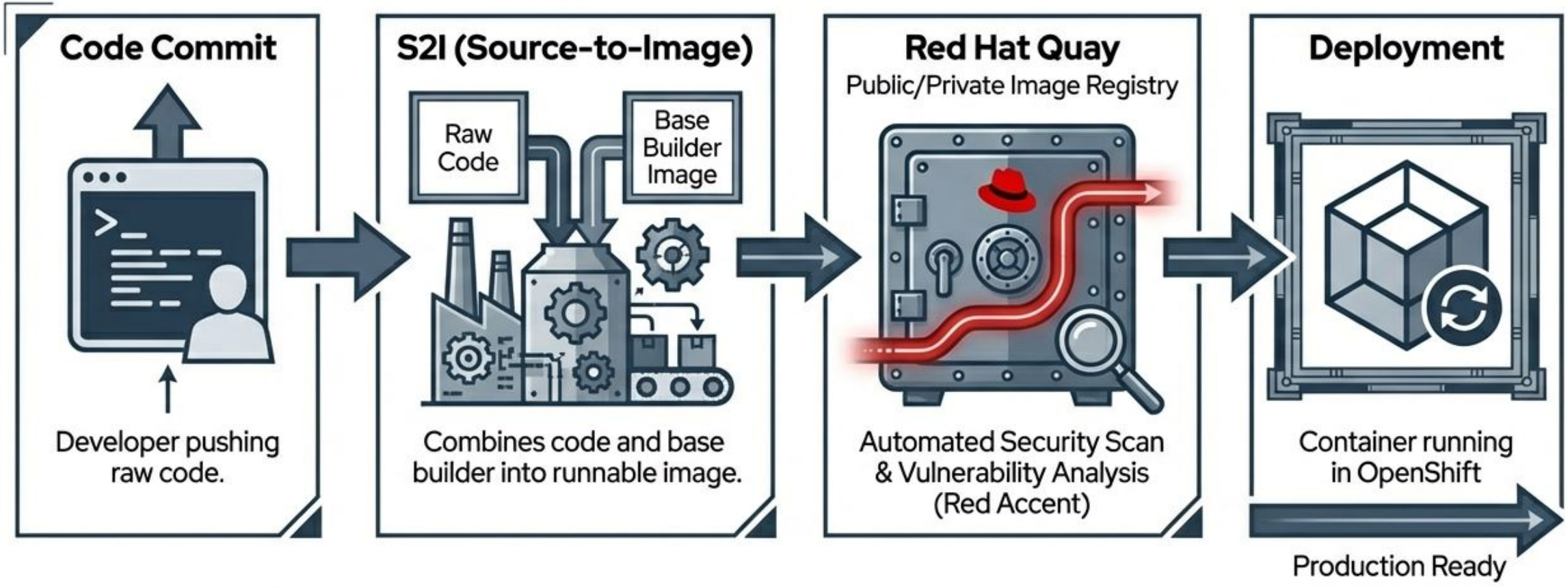
Image Strategy: Source-to-Image (S2I) & Red Hat Quay

Removing the burden of Dockerfile creation while ensuring enterprise-grade artifact security and availability.



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Developer Toolchain – Ecosystem

- IDE plugins and devfiles (OpenShift Dev Spaces / CodeReady-style flows)
- Service binding and bindable operators for databases and messaging
- Observability: metrics, logging, tracing for application teams
- Image registries and supply chain: signatures, scans, policies
- API management and Kafka/AMQ patterns via Operators

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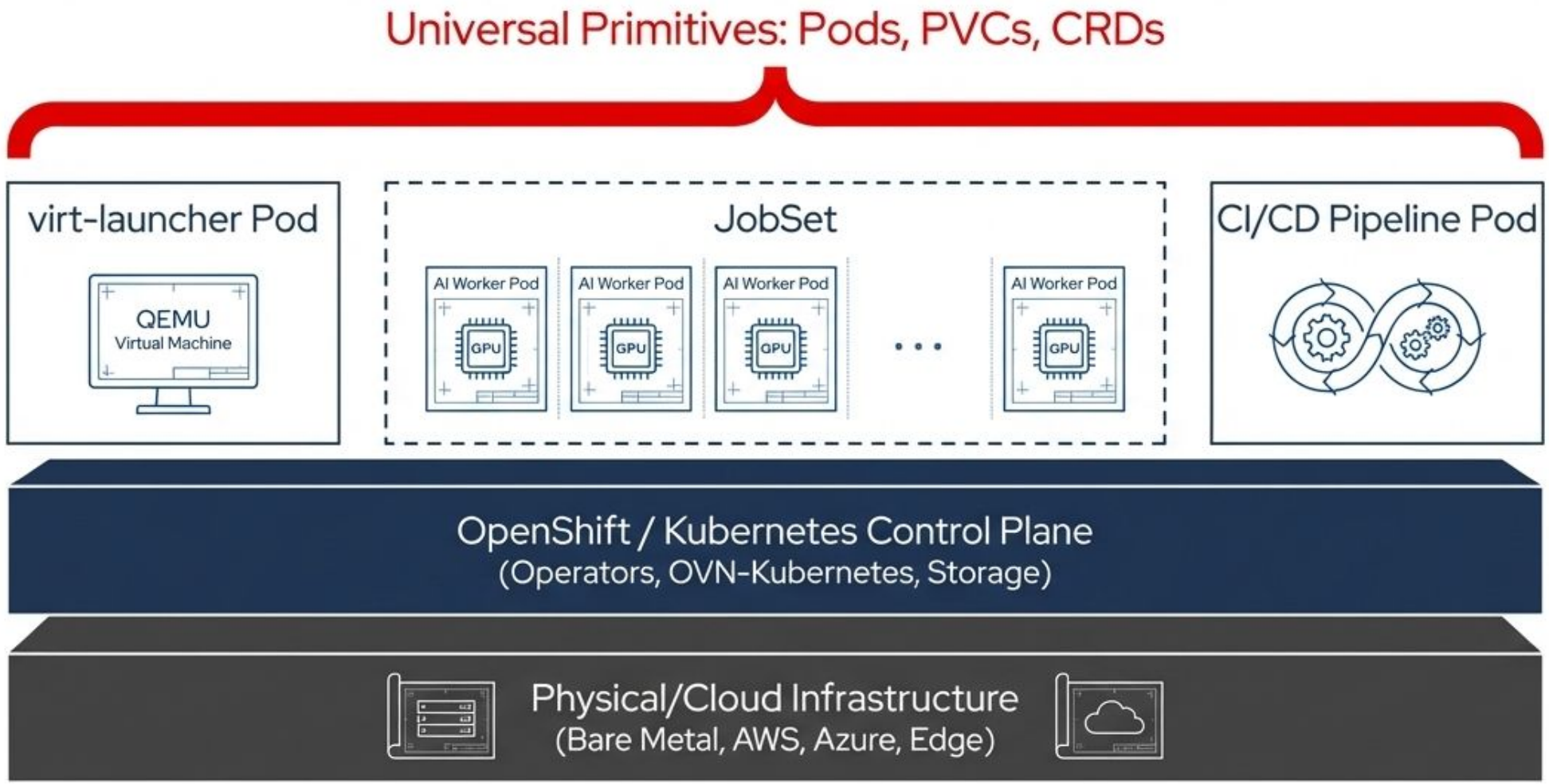
How It Fits Together

Synthesis: The Ultimate Convergence

One platform. One operational model. Infinite workload possibilities.



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Three Pillars at a Glance

High level

Virtualization:
VMs + containers
Legacy + modern
KubeVirt

OpenShift AI:
Models & notebooks
GPU-aware
MLOps patterns

Developer toolchain:
CI/CD & GitOps
Console & oc
Inner/outer loop

Often combined: AI services in containers, VMs for data sources.

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Summary

- OpenShift Virtualization unifies VMs and Kubernetes on one cluster
- OpenShift AI brings ML and GenAI workloads onto the same platform
- The developer toolchain ties Git, builds, pipelines, and GitOps to production
- Together they reduce silos: one operations model, consistent security
- See Red Hat documentation for each product area and version

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Thank You

Questions? See docs.redhat.com – OpenShift Virtualization, OpenShift AI, and Developer guides.

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